UNIT – 1

1. What is Data Science? Mention any two application areas.
2. What are the main stages of a Data Science project?
3. Explain the various stages of a Data Science project.
4. Explain the essential steps involved in data collection and management in a Data Science project.
5. How do you install R for data analysis? Describe the sequence of steps.
6. Describe the steps involved in installing and setting up R and RStudio.
7. Explain the benefits of using IDEs and text editors in R for data analysis.
8. Explain the concept of working with directories in R and why they are essential.
9. What are the basic/common data types available in R?
10. Explain different data types in R with examples.
11. Explain the process of loading and handling data in R.
12. Describe how to load data in R.
13. Write a program in R to read data from an XML file.
14. Explain the process of loading and handling data in R using files and databases.
15. Explain essential R commands and functions used for data exploration and summary statistics.
16. Write any two advantages of using R for data exploration.

UNIT – 2

1. What is a Data Frame in R?
2. Explain the role of Data Frames in R and how they are structured.
3. Write about a Data Frame and explain various operations on it.
4. Describe any two R functions used to handle data in Data Frames.
5. Illustrate the purpose of the `subset()` function in R with Data Frames.
6. Illustrate the process of loading a Data Frame in R with an example.
7. What is the significance of data summary statistics in data analysis, and how can they be generated in R?
8. Relate the purpose of descriptive statistics in exploratory data analysis.
9. Define missing values and invalid values and give an example for each.
10. Explain the concept of finding missing values.
11. Explain different methods to handle missing values in R with examples.
12. Write a short note on invalid values and outliers.
13. Define an outlier. Explain the procedure of identifying outliers in a Data Frame using R.
14. Explain how data visualization helps in spotting problems in data.
15. “Data visualization helps in identifying and addressing data problems.” Justify the statement.
16. Describe the steps involved in cleaning data in R, including strategies for handling missing and invalid values.
17. Differentiate between data cleaning and data validation. Explain data cleaning techniques.
18. Describe the steps for cleaning and preparing data for modeling and validation.

19. Explain the purpose of sampling in the context of data modeling and validation.
20. Write R commands to load a CSV file and display its first 10 rows.
21. Mention one method to handle missing values in data.

UNIT – 3

1. What is memorization in Machine Learning?
2. How do you map real-world problems to appropriate machine learning tasks?
3. Analyze the factors that should be considered while mapping real-world problems to machine learning tasks.
4. Explain the process of choosing, evaluating, and validating models in Data Science.
5. What is the purpose of evaluating models in Machine Learning?
6. Mention common evaluation metrics used to assess the performance of Machine Learning models.
7. Summarize the common evaluation metrics used for assessing Machine Learning models.
8. How do you validate a Machine Learning model to ensure its reliability?
9. Describe the methods used for validating models.
10. Discuss the importance of model validation and different validation techniques.
11. Why are training data and validation data different?
12. What is KDD (Knowledge Discovery in Databases)?
13. What is the significance of KDD in data mining and Machine Learning?
14. Describe the KDD process with reference to knowledge discovery from data.
15. Explain KDD Cup 2009 in Machine Learning and its role in model building.
16. Describe the main objective of the KDD Cup 2009 conference.
17. Write a short note on the contributions of KDD Cup 2009.
18. Describe the procedure of building single-variable models with a suitable example.
19. Describe building models using a single variable and explain why more variables are added to a model.
20. Differentiate between single-variable and multivariable models with examples.
21. Discuss the key advantage of building models using many variables in Machine Learning.

UNIT – 4

1. What is the primary objective of Linear Regression in data analysis?
2. Explain Linear Regression.
3. Compare and contrast simple Linear Regression and Multiple Linear Regression.
4. Explain the process of building a Linear Regression model.
5. Explain how predictions are made using Linear Regression.
6. What is Logistic Regression?
7. What type of outcomes does Logistic Regression predict?
8. What is the primary purpose of Logistic Regression?
9. Differentiate between Linear Regression and Logistic Regression.
10. Describe Logistic Regression and its advantages over Linear Regression.
11. Explain the Generalized Linear Model (GLM).
12. Describe the process of building and interpreting a Logistic Regression model in R.
13. Explain supervised and unsupervised methods with examples.

14. What is cluster analysis?
15. Explain the primary goal of cluster analysis in data mining.
16. How do you choose an appropriate distance metric for a given clustering problem?
17. Outline the key steps of data preparation for clustering analysis.
18. Explain K-Means clustering with an example.
19. Illustrate the working principle of the K-Means algorithm with an example.
20. Discuss the steps of K-Means clustering with an example.
21. What is the key principle behind the K-Means algorithm?
22. How is the number of clusters determined in K-Means?
23. Mention any two disadvantages of the K-Means algorithm.
24. State one real-world application of K-Means clustering.
25. Explain how Hierarchical Clustering works.
26. Define Hierarchical Clustering and mention its use.
27. Explain the main advantage of Hierarchical Clustering in cluster analysis.
28. "Hierarchical Clustering is advantageous in cluster analysis." Justify the statement with an example.

UNIT – 5

1. What is Time Series data?
2. What are the components of Time Series data and what is their significance?
3. Mention some common applications of Time Series analysis in real-world scenarios.
4. What is Time Series similarity search?
5. What is the goal of similarity search in Time Series data analysis?
6. Explain regression and trend analysis in Time Series data.
7. Discuss the basic principles and applications of regression and trend analysis in understanding and modeling time-dependent patterns.
8. How can you read Time Series data in R? Explain with an example.
9. List the common file formats for storing Time Series data and explain how to read them into R.
10. Explain the process of plotting Time Series data in R.
11. Explain the process of plotting and analyzing Time Series data in R.
12. What is the basic need for decomposing Time Series data?
13. Explain Time Series decomposition with a suitable example.
14. Describe the steps involved in decomposing Time Series data into seasonal and non-seasonal components.
15. Describe how Time Series data can be decomposed using R.
16. Explain the importance of understanding the residual component in Time Series analysis.
17. What is seasonal adjustment? Give an example.
18. Which method is commonly used for making forecasts in Time Series analysis in the presence of trend and seasonality?
19. Explain forecasting using Exponential Smoothing.
20. What are Non-Seasonal ARIMA models and Seasonal ARIMA models?
21. Explain ARIMA models with a real-world example.
22. Discuss ARIMA models for Time Series forecasting.

23. Apply ARIMA models in any one real-world application.